
What correction makes a cached computation behave like the skipped full computation?

Paper explainer · arXiv:2606.26778 · ICML 2026 · an independent redrawing of the mechanism, not a reproduction of the paper's figures

LearnBridge

Huang, Chen, Shen & Zhang · ICML 2026

Feature caching buys speed by skipping blocks

A Diffusion Transformer evaluates a deep network at every denoising step. Feature caching reuses a representation computed at an earlier step so later steps can skip expensive blocks.

WHY IT IS HARD

- The reused feature is stale. It was computed for a different timestep, so it carries the wrong content.
- The same approximation is applied at every skipped step, and the error compounds along the trajectory.
- The problem grows exactly where the win is. The longer the cache interval, the larger the drift and the larger the speedup.

CONSTRAINTS

- The backbone stays as it is. No architectural change and no retraining of the base model.
- The calibration budget must be small, a handful of prompts rather than a finetune.
- Skipped steps must stay cheap, otherwise the speedup is spent on the fix.

Existing methods forecast from history

A. INTUITION

Nearby steps produce similar features, so an earlier feature stands in well.

B. MECHANISM

Cache the block output at a reference step, then approximate skipped steps from history: reuse (DeepCache, FORA) or Taylor extrapolation (TaylorSeer).

C. WHERE IT FAILS

- A fixed rule cannot track how the representation changes away from the reference step.
- TaylorSeer assumes smooth, high-order differentiable trajectories.
- At high acceleration the error accumulates.

RELATION TO US

Keep the schedule; replace the rule with a learned, structured correction.

Change the question, learn the correction

THE USUAL QUESTION

Predict the future feature from historical features with a fixed rule, and hope the prediction is close.

LEARNIBRIDGE ASKS

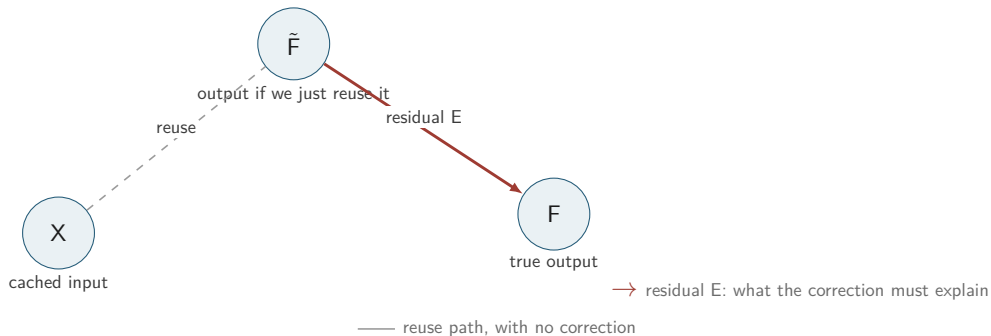
What correction is required for the cached computation to reproduce the skipped full computation?

THE IDEA

Model the required change directly and constrain it. The cached input times a low-rank correction approximates the future full-computation output.

If the required correction has exploitable structure, a few calibration prompts are enough and the correction is nearly free at inference.

What correction is actually required



Over many prompts the paper writes the residuals as a matrix E and the cached inputs as a matrix X , and models $E \approx \Delta W \cdot X$. The geometry above is in feature space; the correction ΔW itself acts on layer weights.

Diagnostic D1: is the required correction low-rank?

D1 Does the optimal calibration update have exploitable structure?

tests C1

MEASUREMENT

Singular values of X (100 prompts) decay rapidly. Fit $E \approx \Delta W \cdot X$: $\Delta W = E \cdot \text{pinv}(X)$, $\text{rank}(\Delta W) \leq \text{rank}(X) \leq r$

OBSERVATION

Spectral energy concentrates in a few principal directions, so the least-squares correction is low rank.

INTERPRETATION

A low-rank parameterization can represent the required correction; no full-rank update is needed.

CAN CONCLUDE

Under the paper's model $E \approx \Delta W \cdot X$, the optimal ΔW is low-rank in the authors' setting.

CANNOT CONCLUDE

That diffusion features are low-rank in general, or that the bound holds for every layer and model.

ALTERNATIVE TARGETED

The decay may partly reflect the chosen layer and prompt pool; this is empirical, not universal.

Diagnostic D2: is the correction shared across prompts?

D2 Does the required correction depend on the prompt?

tests C1

MEASUREMENT

Fit ΔW on each of 20 disjoint prompt groups; compare principal subspaces pairwise: the angles are small.

OBSERVATION

Different prompt groups need nearly the same correction subspace.

INTERPRETATION

The pattern is stable enough that calibration from a few prompts transfers.

CAN CONCLUDE

In the tested setting, subspaces from disjoint prompt groups coincide.

CANNOT CONCLUDE

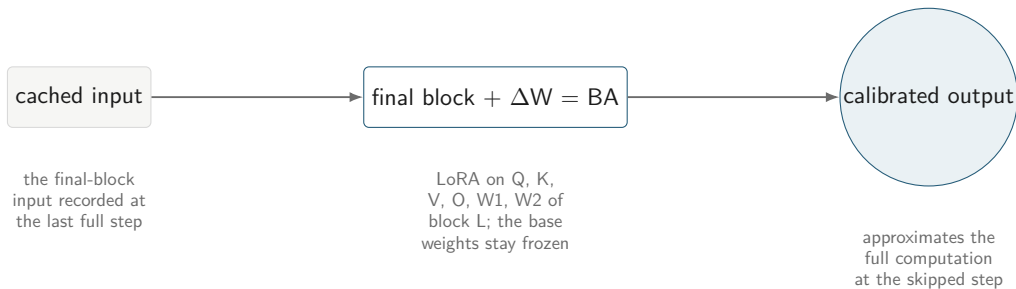
That every prompt needs the same correction, or that invariance survives a domain shift.

ALTERNATIVE TARGETED

The groups came from one prompt pool, which may inflate their similarity.

Huang et al., ICML 2026, Figure 2

Low rank suggests a LoRA bridge on the final block



Low rank is not an extra constraint here: LoRA is exactly a low-rank parameterization, $\Delta W = BA$ with r far below the layer dimensions. The only trainable parameters are the two small factors A and B .

Calibration: one full pass, then only the adapters train

A. PRE-CALIBRATION PASS (FULL COMPUTATION)

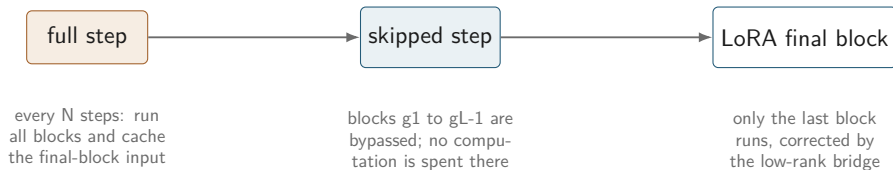
- Record input. At each full-computation step t , keep the final-block input.
- Record target. At the steps that will be skipped, keep the true final-block output.

B. LORA FINETUNE (ONLY THE ADAPTERS TRAIN)

The LoRA-augmented final block must turn the cached input into that recorded target, while every base weight stays frozen.

Budget reported in the paper: 3-5 calibration prompts, small rank, no backbone finetuning.

Inference: skip the blocks, run the bridge



This is the whole acceleration: full computation every N steps, one calibrated block in between. The bridge is not run through every block.

Text-to-image: FLUX.1-dev at about 6x

Method	Speed	ImageReward	CLIP	PSNR	SSIM
ToCA (N=12)	5.87	0.70	0.78	26.48	0.59
TeaCache (delta=1.4)	5.99	0.73	0.80	26.58	0.63
TaylorSeer (N=8, O=2)	6.10	0.82	0.80	26.82	0.68
LearniBridge (N=8) _(ours)	6.17	0.83	0.82	28.35	0.69

higher is better 200 DrawBench prompts. At a slightly higher speedup, LearniBridge keeps the best ImageReward, CLIP, PSNR and SSIM of this group. PSNR and SSIM measure fidelity to the unaccelerated output, not perceptual quality on their own.

Huang et al., ICML 2026, Table 1

Text-to-video: WAN 2.1-1.3B on VBench

Method	Speed	VBench	PSNR	SSIM
EasyCache (delta=0.13)	3.27	79.61	13.77	0.47
TaylorSeer (N=4, O=2)	3.46	79.17	14.84	0.45
TeaCache (delta=0.3)	3.98	79.23	12.82	0.47
LearniBridge (N=5) _(ours)	4.10	80.51	16.32	0.63
LearniBridge (N=4) _(ours)	3.49	81.21	21.26	0.82

higher is better 946 prompts, five samples each. At 4.10x the bridge keeps VBench 1.28 points above the best baseline at a comparable ratio, and at 3.49x it stays close to the unaccelerated model. HunyuanVideo shows the same pattern up to 5.75x in the paper's Table 2.

Huang et al., ICML 2026, Table 3

Our runs: N=5 and N=7 (CLIP, IR, HPSv2.1, PSNR, SSIM, speed)

Method	CLIP	IR	HPSv2.1	PSNR	SSIM	Speed
Original FLUX	32.65	1.00	30.40	∞	1	1.00x
Cache thường (N=5)	32.63	0.94	29.44	14.70	0.55	4.63x
Global-Local (N=5)	32.79	1.01	30.00	15.21	0.57	4.60x
LearnIBridge-100 (N=5) _(paper method)	32.66	0.98	29.53	15.07	0.55	4.49x
Cache thường (N=7)	32.28	0.83	28.24	14.14	0.52	5.66x
Global-Local (N=7)	32.63	0.92	29.10	14.73	0.55	5.60x
LearnIBridge-100 (N=7) _(paper method)	32.76	0.92	28.44	14.49	0.54	5.45x

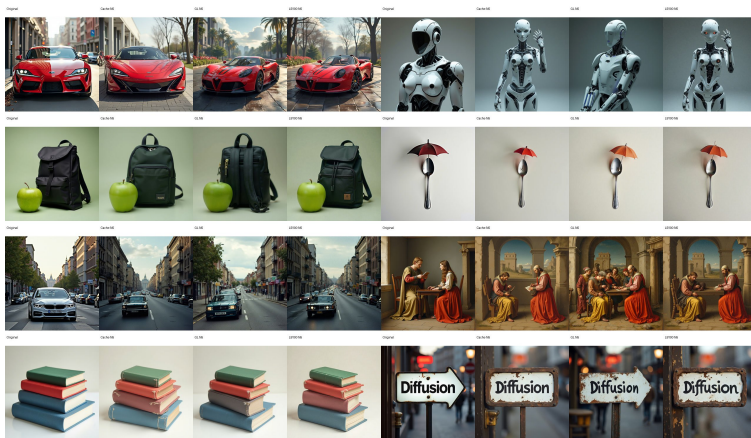
higher is better Fidelity metrics PSNR/SSIM compare against the unaccelerated Original FLUX output, so they fall for every accelerator including the best one; CLIP/IR/HPSv2.1 are the quality signals. LPIPS is omitted here for width (it tracks PSNR/SSIM).

Our runs: N=10 and N=25 (quality collapse at long intervals)

Method	CLIP	IR	HPSv2.1	PSNR	SSIM	Speed
Cache thường (N=10)	31.72	0.66	26.36	13.77	0.48	8.48x
Global-Local (N=10)	31.77	0.76	27.76	14.46	0.52	8.32x
LearniBridge-100 (N=10) _(paper method)	32.16	0.73	26.68	14.10	0.47	7.96x
Cache thường (N=25)	23.08	-1.48	13.71	13.12	0.42	16.87x
Global-Local (N=25)	22.99	-1.28	15.68	14.26	0.49	16.24x
LearniBridge-100 (N=25) _(paper method)	23.09	-1.51	13.72	13.47	0.45	14.77x

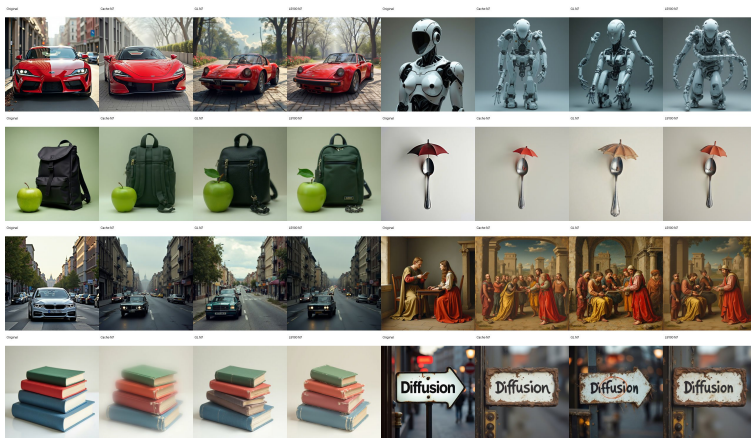
higher is better At 25-step intervals every method collapses (IR below zero, HPSv2.1 near half), so the useful regime in these runs is N=5-10; the qualitative folders show where the remaining failures live.

Qualitative, N=5: Original / Cache / GL / LearniBridge-100



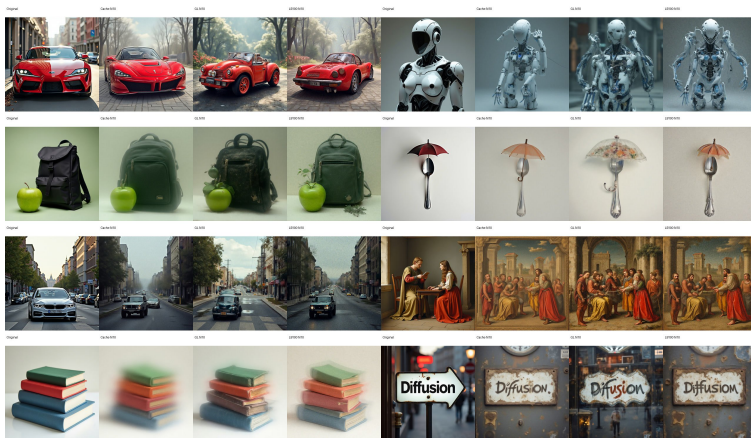
Sheet split in two blocks: rows 1-4 on the left, rows 5-8 on the right. Every row is one prompt, columns are Original, Cache (N=5), GL, LearniBridge-100.

Qualitative, N=7: Original / Cache / GL / LearniBridge-100



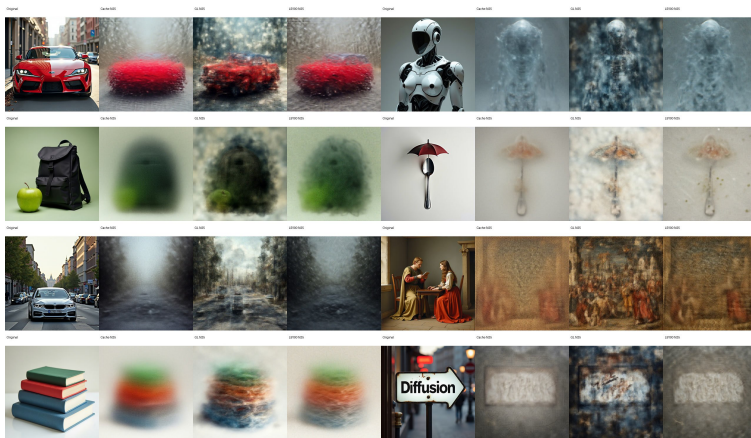
Sheet split in two blocks: rows 1-4 on the left, rows 5-8 on the right. Every row is one prompt, columns are Original, Cache (N=7), GL, LearniBridge-100.

Qualitative, N=10: Original / Cache / GL / LearniBridge-100



Sheet split in two blocks: rows 1-4 on the left, rows 5-8 on the right. Every row is one prompt, columns are Original, Cache (N=10), GL, LearniBridge-100.

Qualitative, N=25: Original / Cache / GL / LearniBridge-100



Sheet split in two blocks: rows 1-4 on the left, rows 5-8 on the right. Every row is one prompt, columns are Original, Cache (N=25), GL, LearniBridge-100.

Limitations and open questions

CURRENT LIMITATIONS

ct Object

- Only the final Transformer block is calibrated, so the bridge has to absorb the shift of every bypassed block.
- Rank is a design choice. The ablation peaks around $r=64$ and degrades at larger ranks.
- The low-rank and prompt-invariance findings are empirical measurements on the tested models and prompt sets.

OPEN QUESTIONS

- Does the shared low-rank structure survive other backbones, modalities and layer choices?
- Can the pre-calibration pass be replaced by data collected during normal inference?
- How does the bridge combine with step distillation, quantization or other caching schedules?